# NOTRE DAME UNIVERSITY BANGLADESH

# Computer Networks Lab Report-01

**Course Code: CSE-3204**
**Course Title: Computer Networks Lab**
**Lab Task Topic: Configuration of Dynamic NAT**

## Submitted by:

**Name: Istiak Alam**
**ID: 0692230005101005**
**Batch: CSE-20**

**Submission Date:** April 22, 2025

## Submitted to:

**Dr. Fernaz Narin Nur**
**Adjunct Professor, NDUB**

# Objective

To implement and understand the working of **Dynamic NAT (Network Address Translation)** using Cisco Packet Tracer, allowing private IP addresses to access a public network by translating them dynamically to a public IP pool.

# Tools Used

- Cisco Packet Tracer
- Routers (2x: Public Router & NAT Router)
- Switches (1 or more)
- End Devices (PC, Server)
- Copper Cross-Over & Serial DCE cables

# Network Design Overview

| **Device** | **Interface** | **IP Address / Role** |
|---|---|---|
| PC | NIC | 192.168.30.2 (Internal User Device) |
| Public Router | Fa0/0 | 192.168.30.1 (Gateway to NAT Router) |
| Public Router | S2/0 | 20.0.0.2 (Link to NAT Router) |
| NAT Router | S2/0 | 20.0.0.1 (NAT Inside) |
| NAT Router | Fa0/0 | 10.0.0.1 (NAT Outside) |
| Server | NIC | 10.0.0.100 (External Server) |

# Step by Step Configuration

## 0.1 PC Configuration

```
IP Address: 192.168.30.2, 192.168.30.3 & 192.168.30.4
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.30.1
```

# Network Topology in Packet Tracer :

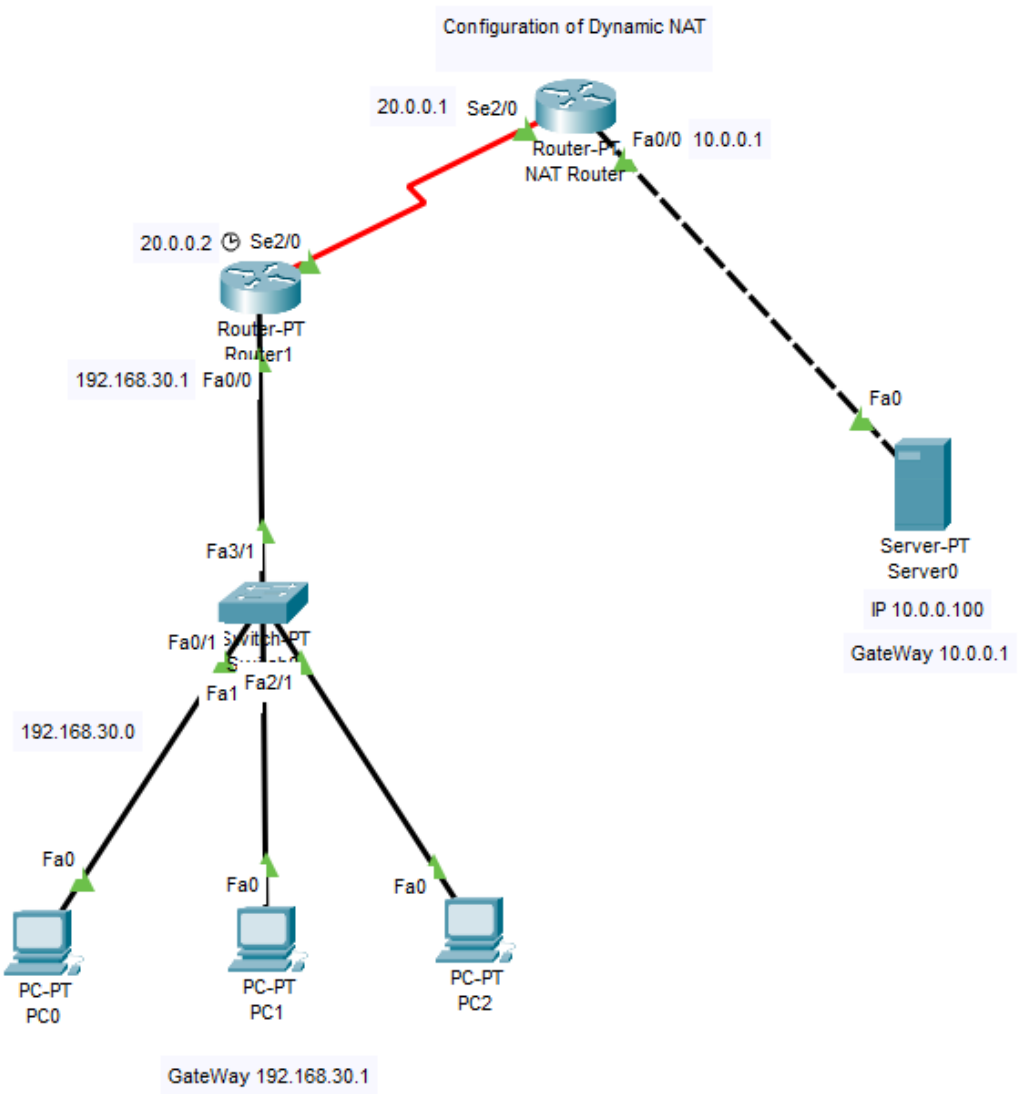


Figure 1: Network Topology for Dynamic NAT Configuration

## 0.2 Public Router Configuration

```
enable
configure terminal

interface fa0/0
ip address 192.168.30.1 255.255.255.0
no shutdown
exit

interface s2/0
ip address 20.0.0.2 255.255.255.252
clock rate 64000
no shutdown
exit

ip route 0.0.0.0 0.0.0.0 20.0.0.1
```

## 0.3 NAT Router Configuration

```
enable
configure terminal

interface s2/0
ip address 20.0.0.1 255.255.255.252
ip nat inside
no shutdown
exit

interface fa0/0
ip address 10.0.0.1 255.255.255.0
ip nat outside
no shutdown
exit

ip nat pool DYN_POOL 10.0.0.20 10.0.0.30 netmask 255.255.255.0
access-list 1 permit 192.168.30.0 0.0.0.255
ip nat inside source list 1 pool DYN_POOL overload

ip route 192.168.30.0 255.255.255.0 20.0.0.2
```

## 0.4 Server Configuration

```
IP Address: 10.0.0.100
Subnet Mask: 255.255.255.0
Default Gateway: 10.0.0.1
```

# Testing the Configuration

- From PC, run: `ping 10.0.0.100`
- On NAT Router, verify NAT translations:

```
show ip nat translations
```

Output:

```
Pro  Inside Global     Inside Local     Outside Local     Outside Global
icmp 10.0.0.20:6       192.168.30.3:6     10.0.0.100:6       10.0.0.100:6
icmp 10.0.0.20:7       192.168.30.3:7     10.0.0.100:7       10.0.0.100:7
icmp 10.0.0.20:8       192.168.30.3:8     10.0.0.100:8       10.0.0.100:8
icmp 10.0.0.20:9       192.168.30.3:9     10.0.0.100:9       10.0.0.100:9
```

# Conclusion

This lab successfully demonstrated how **Dynamic NAT** works in a network, allowing internal private IPs to dynamically be assigned public IPs from a pool when accessing an external network. The configuration ensures secure and controlled access between internal and external networks.